Horizontal and vertical dilations

1

a 360

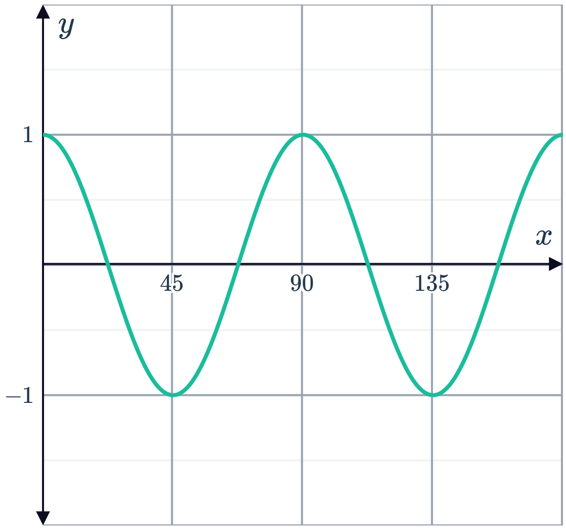
b

x	0	22.5	45	67.5	90	112.5	135	157.5	180
$g(x)$	1	0	-1	0	1	0	-1	0	1

c 90

d Horizontal dilation by a factor of $\frac{1}{4}$

e

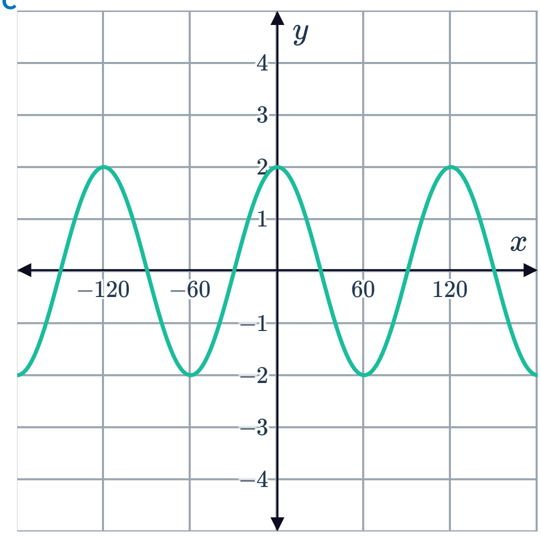


2

a 2

b 120

c



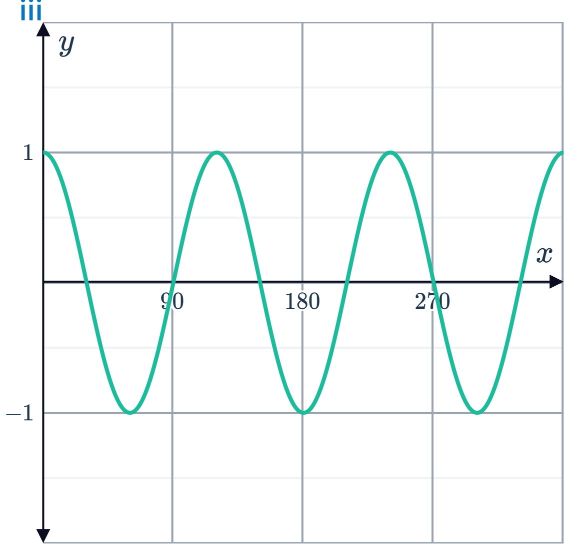
3

a

i $A = 1$

ii $P = 120$

iii

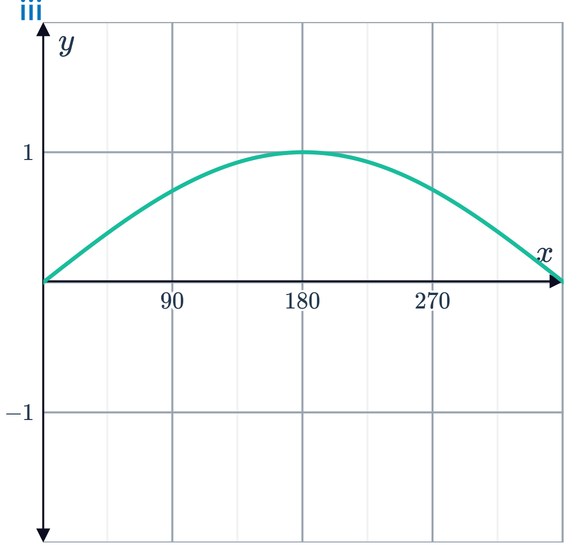


b

i $A = 1$

ii $P = 720$

iii

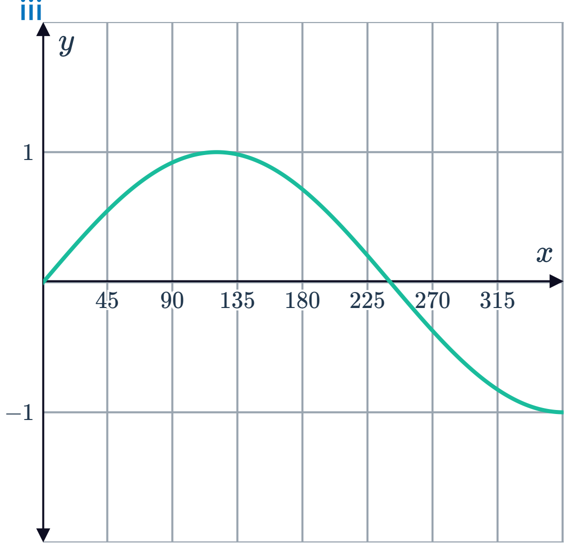


c

i $A = 1$

ii $P = 480$

iii

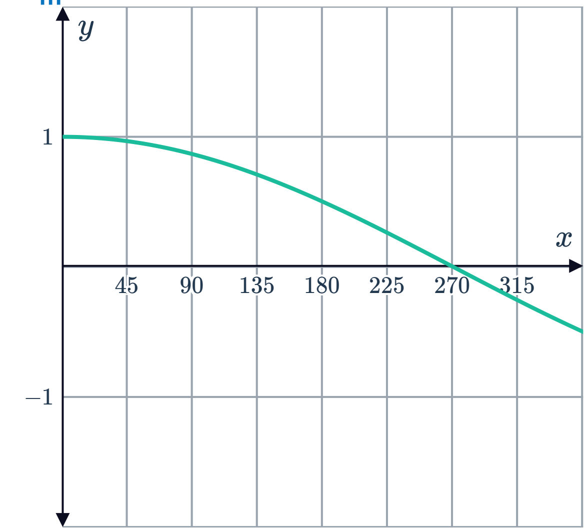


d

i $A = 1$

ii $P = 1080$

iii

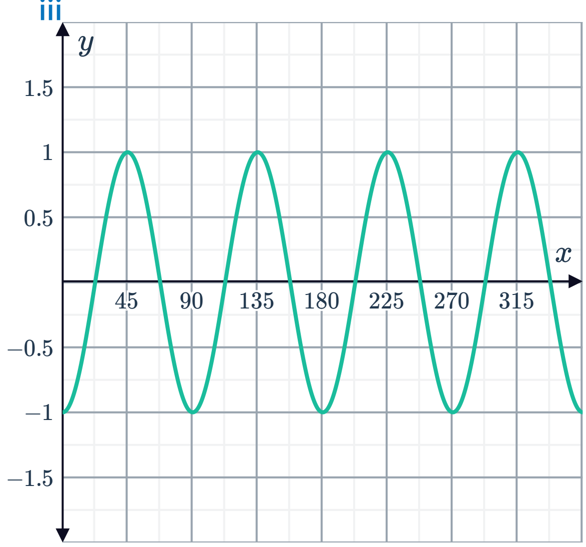


e

i $A = 1$

ii $P = 90$

iii

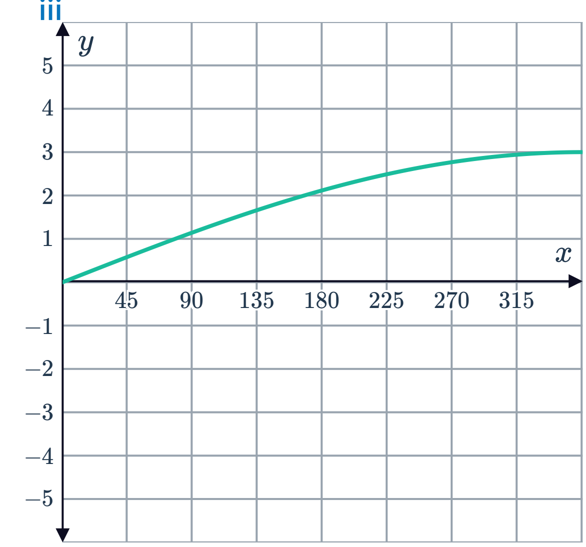


f

i $A = 3$

ii $P = 1440$

iii

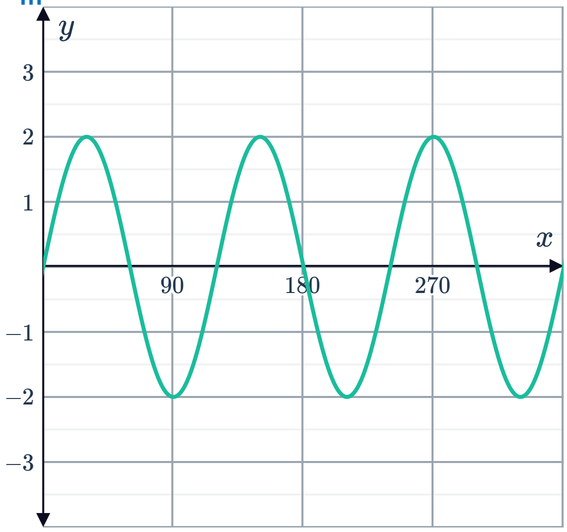


g

i $A = 2$

ii $P = 120$

iii

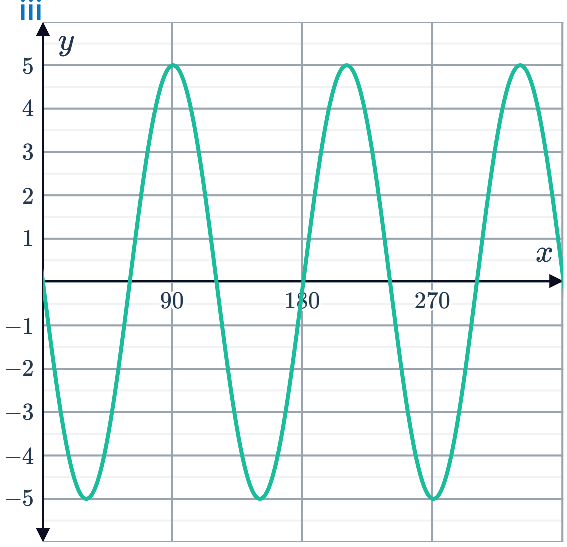


h

i $A = 5$

ii $P = 120$

iii

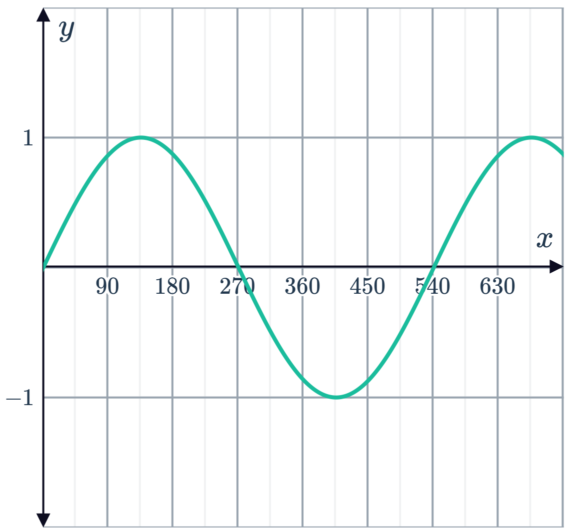


4

a 1

b 540°

c



5

a

x	0	360	720	1080	1440
$\sin\left(\frac{x}{4}\right)$	0	1	0	-1	0

b 1440

6

a Horizontal dilation by a factor of $\frac{1}{4}$.

b 4

Combined translations and dilations

7

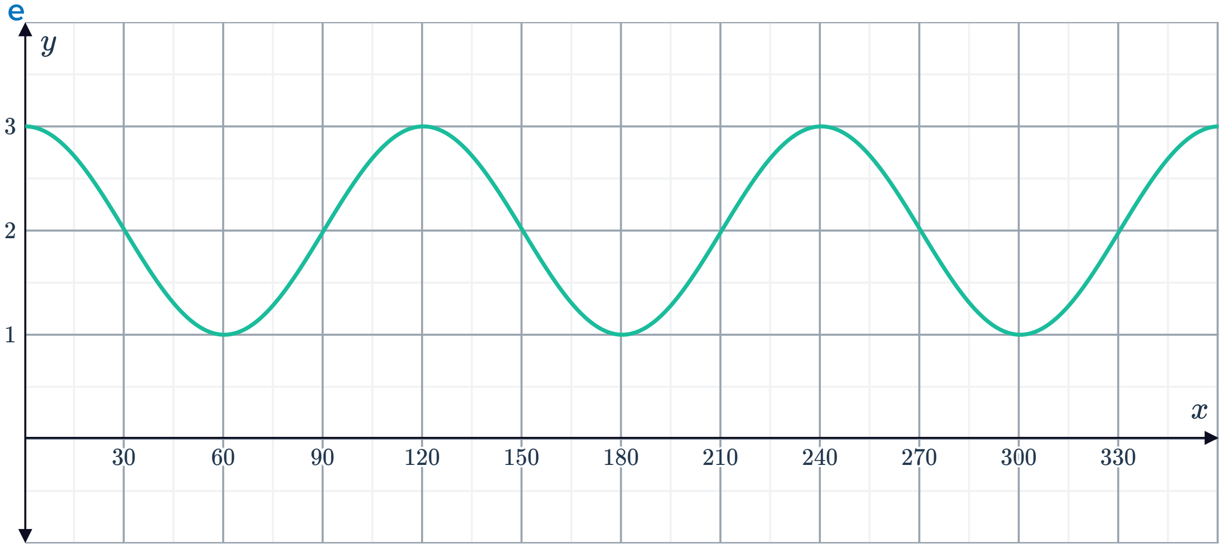
a 120

b 1

c 3

d 1

e



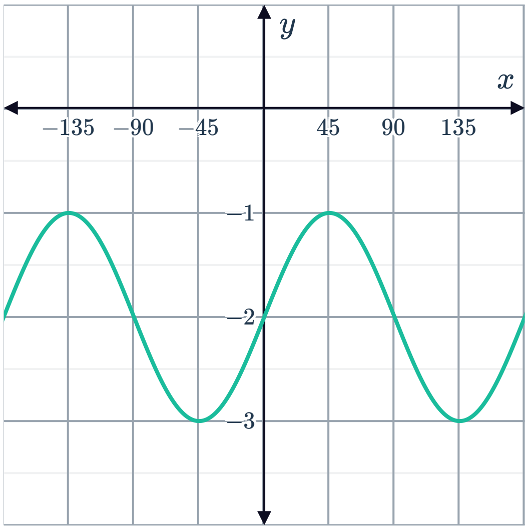
8

a

i $(-\infty, \infty)$

ii $[-3, -1]$

iii

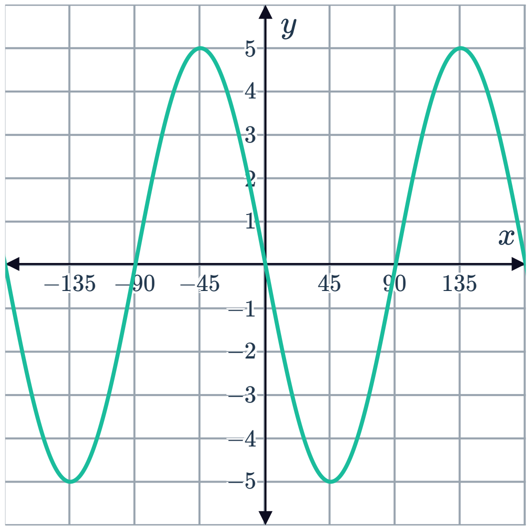


b

i $(-\infty, \infty)$

ii $[-5, 5]$

iii

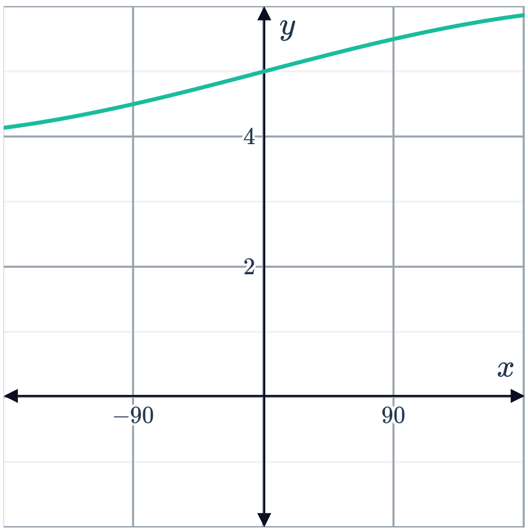


c

i $(-\infty, \infty)$

ii $[4, 6]$

iii

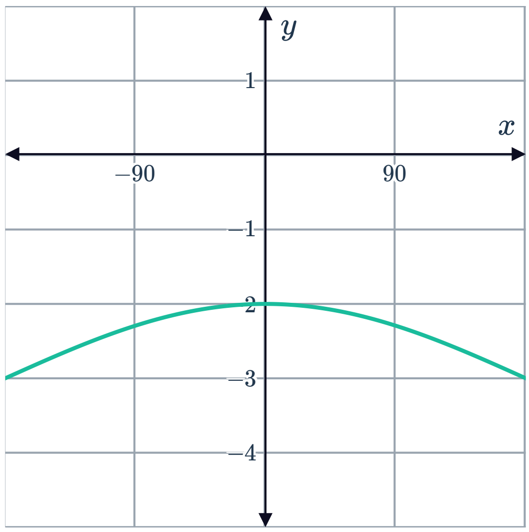


d

i $(-\infty, \infty)$

ii $[-4, -2]$

iii



9

a $y = 3\sin 2x$

b $y = -\sin 2x$

c $y = 2\cos 3x$

d $y = -4\cos\left(\dfrac{x}{2}\right)$

10

a Yes

b No

c Yes

d No